

DESAFIO



Sistema de Arquivos

100

John é um especialista em sistemas de arquivos e estava profundamente envolvido em um projeto confidencial. Utilizando editores hex, ele estava minuciosamente analisando um arquivo crucial.

No entanto, um atacante astuto conseguiu obter um dump de memória do sistema de John.

Sua missão é analisar o dump para descobrir o tamanho exato do arquivo excluído que John estava estudando, pois o futuro do projeto depende dessa informação!

Máscara: FIAP{xxxxxxbytes}



PONTUAÇÃO BÔNUS PARA OS PRIMEIROS A RESOLVEREM



+70 pts



+50 pts



+30 pts



memdump.mem...

Bandeira

Enviar

Identificado o profile:

```
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
INFO    : volatility.debug      : Determining profile based on KDBG search...
        Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
        AS Layer1           : IA32PagedMemoryPae (Kernel AS)
        AS Layer2           : FileAddressSpace (/mnt/f/Downloads/New Folder/memdump.mem)
        PAE type            : PAE
        DTB                 : 0xb40000L
        KDBG                : 0x80545ae0L
        Number of Processors : 1
        Image Type (Service Pack) : 3
        KPCR for CPU 0       : 0xffdff000L
        KUSER_SHARED_DATA    : 0xffdf0000L
        Image date and time  : 2023-10-14 15:18:32 UTC+0000
        Image local date and time : 2023-10-14 12:18:32 -0300
```

Utilizado o pslist que é para listar os processos que estavam em execução no sistema no momento da captura.

Offset(V)	Name	PID	PPID	Thds	Hnds	Sess	Wow64	Start	Exit
0x817cc830	System	4	0	59	251	-----	0		
0x816a4970	smss.exe	376	4	3	19	-----	0	2023-10-14 15:16:48 UTC+0000	
0x81501da0	csrss.exe	528	376	10	440	0	0	2023-10-14 15:16:48 UTC+0000	
0x8144cc78	winlogon.exe	556	376	26	469	0	0	2023-10-14 15:16:48 UTC+0000	
0x81497da0	services.exe	668	556	15	279	0	0	2023-10-14 15:16:49 UTC+0000	
0x8138a398	lsass.exe	680	556	26	368	0	0	2023-10-14 15:16:49 UTC+0000	
0x813fd3b0	vmacthlp.exe	836	668	1	25	0	0	2023-10-14 15:16:49 UTC+0000	
0x8159a8d8	svchost.exe	848	668	19	199	0	0	2023-10-14 15:16:49 UTC+0000	
0x81663868	svchost.exe	932	668	9	258	0	0	2023-10-14 15:16:49 UTC+0000	
0x814c4710	svchost.exe	1024	668	67	1221	0	0	2023-10-14 15:16:49 UTC+0000	
0x8147a020	svchost.exe	1084	668	6	69	0	0	2023-10-14 15:16:49 UTC+0000	
0x816062e0	svchost.exe	1132	668	14	195	0	0	2023-10-14 15:16:49 UTC+0000	
0x813a4228	spoolsv.exe	1400	668	15	144	0	0	2023-10-14 15:16:51 UTC+0000	
0x80f367e0	explorer.exe	1728	1668	19	383	0	0	2023-10-14 15:16:56 UTC+0000	
0x815d66f8	rundll32.exe	1828	1728	4	76	0	0	2023-10-14 15:16:56 UTC+0000	
0x815d6200	vmtoolsd.exe	1836	1728	7	224	0	0	2023-10-14 15:16:56 UTC+0000	
0x815d5da0	ctfmon.exe	1844	1728	1	69	0	0	2023-10-14 15:16:56 UTC+0000	
0x80f02020	svchost.exe	1992	668	6	88	0	0	2023-10-14 15:17:09 UTC+0000	
0x815dbda0	svchost.exe	236	668	8	126	0	0	2023-10-14 15:17:09 UTC+0000	
0x815b4978	VGAAuthService.e	476	668	2	60	0	0	2023-10-14 15:17:17 UTC+0000	
0x815e6568	vmtoolsd.exe	508	668	9	273	0	0	2023-10-14 15:17:17 UTC+0000	
0x80eae3c70	wmiiprvse.exe	1056	848	13	236	0	0	2023-10-14 15:17:18 UTC+0000	
0x80ecc8b0	TPAutoConnSvc.e	1508	668	5	106	0	0	2023-10-14 15:17:18 UTC+0000	
0x80ebd810	alg.exe	1200	668	7	104	0	0	2023-10-14 15:17:19 UTC+0000	
0x80eada0	TPAutoConnect.e	1548	1508	1	78	0	0	2023-10-14 15:17:20 UTC+0000	
0x80eae8a8	wscntfy.exe	1552	1024	1	37	0	0	2023-10-14 15:17:20 UTC+0000	
0x816fc718	wuauclt.exe	1212	1024	8	175	0	0	2023-10-14 15:18:02 UTC+0000	
0x80eadda0	wuauclt.exe	1784	1024	5	141	0	0	2023-10-14 15:18:17 UTC+0000	
0x814908b0	FTK Imager.exe	1296	1728	10	258	0	0	2023-10-14 15:18:18 UTC+0000	

E vemos o que chama atenção:

0x814908b0	FTK Imager.exe	1296	1728	10	258	0	0	2023-10-14 15:18:18 UTC+0000	
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Depois disso, procurei arquivos associados ao administrador da maquina.

Analisando os handles, filtrando por arquivos.

```
python2 vol.py -f ~/mnt/ff/Downloads/New Folder/memdump.mem --profile=WinXSP3x86 handles -p 1296 | grep -iE "File|Device"
```

Volatility Foundation Volatility Framework 2.6.1

0x80e0ecf8	1296	0xc	0x100020	File	\Device\HarddiskVolume1\Documents and Settings\Administrador\Desktop
0x8172d968	1296	0x10	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x81606c38	1296	0x1c	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.MFC.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_a57c1f53
0x813dc998	1296	0x20	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x8139f180	1296	0x24	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x8139f3e8	1296	0x28	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x8147cee8	1296	0x2c	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x80f38e58	1296	0x30	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x8164e008	1296	0x34	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x813dc770	1296	0x38	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x813dc3c0	1296	0x3c	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x814b5838	1296	0x40	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x814b5610	1296	0x44	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x8134a6e0	1296	0x48	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.MFC.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_a57c1f53
0x81320b00	1296	0x4c	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.MFC.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_a57c1f53
0x814b5a70	1296	0x50	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x8159f630	1296	0x58	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.VC90.CRT.1fc8b3b9a1e18e3b_9.0.30729.4148_x-ww_d495ac4e
0x815e2690	1296	0x7c	0x100001	File	\Device\KsecDD
0x81273028	1296	0x98	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
0x81467000	1296	0xa4	0x100003	Event	userenv: User Profile Setup event
0x8149a210	1296	0x1c4	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
0x8149a688	1296	0x1e0	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
0x8150bc08	1296	0x1e4	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
0x8160b608	1296	0x1fc	0x12019f	File	\Device\HarddiskVolume1\Documents and Settings\Administrador\Configurações locais\Temporary Internet Files\Content.IE5\index.dat
0x80f256a0	1296	0x200	0x100000	Mutant	c:\documents and settings\administrador\configurações locais\temporary internet
Files\content.ie5\					
0xe2338210	1296	0x204	0x2	Section	C:\Documents and Settings\Administrador\Configurações locais\Temporary Internet
Files_Content.IE5_index.dat_32768					
0x81647b58	1296	0x210	0x12019f	File	\Device\HarddiskVolume1\Documents and Settings\Administrador\Cookies\index.dat
0x81647e08	1296	0x218	0x12019f	File	\Device\HarddiskVolume1\Documents and Settings\Administrador\Configurações locais\Histórico\History.IE5\index.dat
0x813a0208	1296	0x254	0x120009	File	\Device\HarddiskVolume1\WINDOWS\system32\shdocvw.dll
0x81458f90	1296	0x25c	0x120089	File	\Device\HarddiskVolume1\WINDOWS\system32\stdole2.tlb
0x81581028	1296	0x264	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
0x813216e8	1296	0x284	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
0x81602bb8	1296	0x288	0x100000	Mutant	RasPbFile
0xe15da0a8	1296	0x2c8	0xf003f	Key	MACHINE\SYSTEM\CONTROLSET001\HARDWARE\PROFILES\0001
0x8139b430	1296	0x2ec	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
0x814b4898	1296	0x328	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
0x81653e78	1296	0x36c	0x12019f	File	\Device\KSENUM\00000001\{98365890-165f-11D0-A195-0020AFD156E4}
0x8139e1f0	1296	0x3a0	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
0x814571c8	1296	0x3a8	0x100020	File	\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
0xe162d108	1296	0x3c0	0xf003f	Key	USER\5-15-21-1229272821-1326574676-1801674531-500\SOFTWARE\MICROSOFT\WINDOWS\CURRENTVERSION\EXPLORER\FILEEXTS
0x81647028	1296	0x3dc	0x12019f	File	\Device\NamedPipe\Wksvc
0x81457e58	1296	0x3e0	0x12019f	File	\Device\NamedPipe\svrsvc
0x80efcb58	1296	0x3f0	0x12019f	File	\Device\NamedPipe\lsarpc
0x8171a770	1296	0x3fc	0x12019f	File	\Device\addriver
0x81464f90	1296	0x400	0x12019f	File	\Device\HarddiskVolume1\Documents and Settings\Administrador\Desktop\memdump.mem
0x81464d90	1296	0x404	0x12019f	File	\Device\addriver

Vemos que tem um arquivo dump de memória.

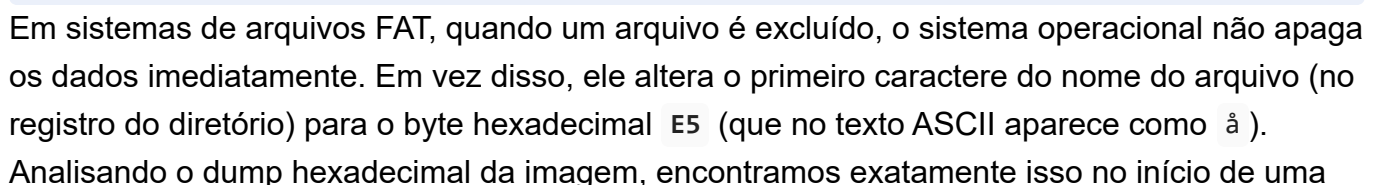
C:\Documents and Settings\Administrador\Desktop\memdump.mem

```
Volatility Foundation Volatility Framework 2.6.1
0x00000000012eecf8      1      0 R--rw- \Device\HarddiskVolume1\Documents and Settings\Administrador\Desktop
0x0000000001189150      1      0 R--rw- \Device\HarddiskVolume1\Documents and Settings\Administrador\Desktop\cmd.png
0x0000000001864f90      1      1 -W-r-- \Device\HarddiskVolume1\Documents and Settings\Administrador\Desktop\memdump.mem
0x00000000019d9ed8      3      1 R--rwd \Device\HarddiskVolume1\Documents and Settings\Administrador\Desktop
```

Vamos recuperar esse arquivo.

Agora precisamos ver o conteúdo de `cmd.png`, porque o nome sugere que ela pode mostrar justamente o comando/procedimento usado para localizar e remontar o arquivo excluído.

bx16496d8.dat - Visualizador de Fotos do Windows



das linhas de registro

E5 4F 54 4F 28 35 7E 31 4A 50 47 (que corresponde a   T  (5~1JPG)).

Em uma entrada de diret  rio FAT de 32 bytes, a arquitetura reserva exatamente os **4   ltimos bytes** (offsets 28 a 31) para registrar o tamanho total do arquivo. Se contarmos os blocos dessa entrada do arquivo E5 at   o final dos 32 bytes, chegamos na   rea imediatamente ap  s a caixa vermelha (  D     , que indica o cluster inicial). Os 4   ltimos bytes desta entrada s  o: 16 27 01    

Agora aplicamos a regra de invers  o de arquitetura que voc   mencionou:

- Lemos os bytes de tr  s para frente:     01 27 16
- Juntamos o valor em hexadecimal:   x012716
- Convertendo o hexadecimal 12716 para decimal:
 - $1 \times 16^4 = 65536$
 - $2 \times 16^3 = 8192$
 - $7 \times 16^2 = 1792$
 - $1 \times 16^1 = 16$
 - $6 \times 16^0 = 6$
 - Soma total: $65536 + 8192 + 1792 + 16 + 6 = 75542$

O tamanho exato do arquivo exclu  do    75.542 bytes.

R:FIAP{75542bytes}